Simple OLS Regression with a Dummy

ECON20222 - Lecture 3

Ralf Becker and Yidi Sun

Recap

- **3 examples:**
  - Minimum Wage (x) and Employment (y)
  - Law Officers per capita (x) and Violent Crime per capita (y)
  - Gun Legislation (x) and Death Rate by Firearms (y)
- **2 datasets:**
  - Card & Krueger (CK)
  - Siegel et al. (2019) (S)
- Some revision econometrics
- One software: R

Structures of Economic Data

- Just like Economics is classified into Macro and Micro, we divide data/econometrics in the same way:
- **Micro-econometric data** :
  - Cross-sectional data
  - Pooled/repeated cross sections
  - Panel or longitudinal data (e.g. CK & S)
- **Macro-econometric data** :
  - Cross-sectional data
  - Time series data
  - Macro panel data

The unit-of-observation is very important.

Micro-econometric Data

- Consist of samples of individuals, households, firms, or other micro-level units. Observations refer to micro-economic units; widely used in development economics, labour economics, and industrial organisation.
- Commonly used to test microeconomic hypotheses and evaluate economic policies.
- Typically collected through: Surveys, Administrative databases, Web-scraped data. Ideally obtained using random sampling methods.
- Observed either:
  - At a single point in time: **cross-sectional data**,
  - Over multiple time periods (usually a few years):
    - When the same micro-units are followed over time: **panel data**
    - When different samples are observed over time: **repeated cross-sections**

Example: US Gun Policy Dataset

- The US Gun Policy dataset from Siegel et al. (2019). The dataset contains:
  - 51 states, indexed by s,
  - 21 years, indexed by t (2001–2021),
  - A total of 1,071 state–year (st) observations,
  - A unit of observation defined as a state–year.
    This is an example of an aggregated (balanced) panel dataset.
- The Card–Krueger dataset is a micro-level (balanced) panel: Two observations for each of 410 fast-food outlets.
- The BHPS/US dataset(next example): Household-level panel dataset.

UK Datasets

Examples of well-used publicly available datasets:

- FES Family Expenditure Survey; Annual survey; About 30,000 different households each year.
- CP Census of Production; Annual survey; Sample of firms aggregated to 4-digit (225) industries.
- BHPS/US British Household Panel Survey; Annual survey; About 10,000 households each year since 1991;The same households are interviewed every year; Since 2009/10 it has been called *Understanding Society*.
- LFS Labour Force Survey; Quarterly rolling panel; About 55,000 households; Each household is interviewed for five consecutive quarters and then dropped.

Much better to look at The UK Data Archive.

Econometrics so far

- Simple regression:

$$y = \alpha + \beta x + u.$$

- The key issue is whether or not the <u>exogeneity assumption</u> holds

$$E(u \mid x) \equiv E(ux) = \mathrm{Cov}(u, x) = 0 \tag{A2}$$

- Even though OLS <u>by construction</u> imposes the sample analogue:

$$\sum_i \hat{u}_i x_i = 0.$$

Reference: Lecture notes - Introduction

Econometrics so far

- Given (A2), the OLS estimator for β is:

$$\hat{\beta} = \frac{\widehat{\mathrm{Cov}}(y,x)}{\widehat{\mathrm{Var}}(x)} = \frac{\sum_i (y_i - \bar{y})(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}.$$

- The data <u>cannot</u> help us decide whether (A2) is true, since u is unobserved.
- If (A2) is true, we can talk about **causal effects** and OLS being an **unbiased estimator**.
- Hypothesis testing was covered twice:

$$H_0 : \beta = 0 \qquad \text{and} \qquad H_0 : \mu_1 = \mu_2.$$

Example 1

Are Danes the happiest nation in the world?

This "stylised fact" is well known, and Danes are often thought to be happier than their neighbours, the Swedes.

You can test this Danes vs. Swedes comparison using methods from Lecture 2:

$$H_0 : \mu_1 = \mu_2 \qquad \text{and} \qquad H_a : \mu_1 \neq \mu_2,$$

using $\bar{y}_1 - \bar{y}_2$ to construct a test statistic.

Example 2

Pfizer–BioNTech Vaccine Efficacy

For microeconomists, the Random Control Trial (RCT) is the gold standard in causal inference. Sometimes, RCTs are important globally. The results of the first Pfizer–BioNTech clinical trial (published December 2020) are:

	Pfizer–BioNTech	Placebo injection
Number in each group	18,198	18,325
Number of Covid-19 cases	8	162
Proportion getting the disease	0.043%	0.884%

The vaccine **efficacy** is 95%.

Example 2

- Taken from Table 22.1 in *Covid by Numbers*. Do not read the original article (in the *New England Journal of Medicine*).
- Efficacy is defined as:

$$E = 1 - \frac{\mu_T}{\mu_C} = \frac{\mu_C - \mu_T}{\mu_C}.$$

- This is a **normalised difference in probabilities**.
- When $\mu_T = 0$ (none of the treated are ill), then $E = 1$ (total efficacy).
- When $\mu_C = \mu_T$, then $E = 0$ (the vaccine has no effect).
- **Vaccine efficacy** is the correct terminology (and is not the same as vaccine **effectiveness**).
- Population probabilities μ are estimated by sample proportions p.

Example 2

- The underlying individual variable y_i is binary:

$$y_i = \begin{cases} 1 & \text{if the outcome is true (here: gets Covid),} \\ 0 & \text{otherwise.} \end{cases}$$

- The reported statistics are sample means, i.e. the proportion of (sub)samples for which $y = 1$: $p = \bar{y} = \frac{1}{n}\sum_i y_i$.
- The data can be summarised by a **cross tabulation**:

	Covid ($y_i = 1$)	**Not Covid** ($y_i = 0$)	
Treated ($d_i = 1$)	n_{11}	n_{01}	$p_T = n_{11}/(n_{11} + n_{01})$
Control ($d_i = 0$)	n_{10}	n_{00}	$p_C = n_{10}/(n_{10} + n_{00})$

Example 2

- In the trial,

$$n_{11} = 8, \quad n_{10} = 168, \quad n_{11} + n_{01} = 18{,}198, \quad n_{10} + n_{00} = 18{,}325,$$

  and the estimated efficacy is calculated as:

$$\hat{E} = 1 - \frac{p_T}{p_C} = \frac{p_C - p_T}{p_C} = 1 - \frac{0.00043}{0.00884} = 0.95.$$

- The reported 95% confidence interval is (0.90, 0.98). (Do not worry about how this is calculated.)

Example 3

How much should tuition fees be?

In the early 2000s, the claim was that graduates, on average, earn more than £400,000 more than non-graduates over their lifetime. This argument was used by the Government at the time to increase tuition fees to £3,200 per year in 2006. After that, fees increased to £9,000 per year following the October 2010 Spending Review (and are now £9,250 per year).

- £400,000 is only an average and varies by subject, occupation, and other characteristics.
- More generally, what is the return to investment in education (schooling)? 5% per annum? 10% per annum?

Example 4

GCSE results

Do girls do better than boys at GCSE? Is this "education gender gap" rising? Did it change with Covid?

Every summer we compare the proportion of girls who *pass* their GCSEs with the proportion for boys:

	2008	2014	2018
Girls	69.4%	73.1%	71.2%
Boys	60.4%	64.3%	62.1%
Gap	9.0%	8.8%	9.1%

See BBC: *"GCSE Grades Rise..."* and
See BBC: *"GCSE results rise despite tougher exams"*.

Example 5

August babies flop at exams

“Amongst girls born in September, 60.7% achieve at least five A* to C grade GCSEs, compared with only 55.2% for those born in August” (IFS).
The corresponding figures for boys are 50.3% and 44.2%.

Is this August/September difference statistically significant?
What information do we need in order to carry out a formal test?

A similar US variation in educational achievement by month of birth was exploited in a famous paper by Joshua Angrist (from the textbook *Mastering 'Metrics*) and Alan Krueger (already seen in Lecture~1).
Angrist shared half of the 2021 Nobel Prize for this paper and related work.
See *Mastering 'Metrics*, Section~6.3.

Example 6

Did the Education Maintenance Allowance (EMA) improve staying-on rates?

In England, between 2004 and 2010, a young person was eligible to receive an allowance of up to £30 per week provided they stayed on at school and their parents earned less than £30,000 per year. The aim of the programme was to encourage more young people to stay on in education.

From the *Longitudinal Survey of Young People in England*, 5,011 young persons were observed in Year~11:

- 2,393 were eligible, of whom 58.1% stayed on.
- 2,618 were ineligible, of whom 63.4% stayed on.

The raw differential is -5.3 percentage points, and so the policy appears <u>not</u> to have worked.

More Examples

Examples

- Is there still gender or race discrimination in the UK labour market?
- Did the introduction of the National Minimum Wage (NMW) in April 1999 affect the employment prospects of low-paid workers?

Can we use the same Difference-in-Differences (DiD) methodology as introduced in Lecture 1 and developed more formally in Lecture 7?

Raw Differentials

- Every week we see evidence like this being reported, mainly coming from micro-econometric data.
- Although we are interested in raw averages themselves, more often we want to compare averages across categories:
  - Danes vs. Swedes
  - Treated vs. control
  - Boys vs. girls
  - (Not) affected by the NMW
  - (In)eligible for EMA
- These comparisons are called **raw differentials**.
- Regression handles this naturally: the RHS variable x becomes a **dummy variable**.
- Differences in raw differentials are also very common and are known as **differences-in-differences** (or **diff-in-diffs**, DiDs).

Simple Regression when x is a Dummy

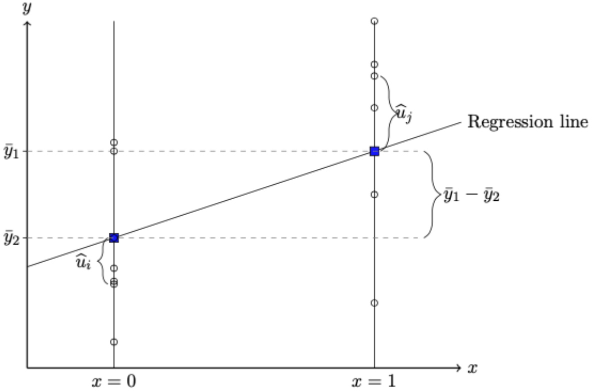
- We start with the same model,

$$y = \alpha + \beta x + u, \qquad x \in \{0, 1\}, \tag{1}$$

  but now x is either 0 (e.g. a woman) or 1 (e.g. a man).

- x is said to be a dummy variable. There are many of these variables in cross-section datasets.

Simple Regression when x is a Dummy



.

Figure 1: A sample of data when x is a dummy. The white circles represent typical data points and the blue rectangles represent sample averages.

Simple Regression when x is a Dummy

- To estimate the two parameters α and β, we require two moment conditions (exogeneity assumptions):

$$E(u \mid x = 0) = E(u \mid x = 1) = 0. \tag{A3}$$

- Write the population regression function for the cases $x = 0$ and $x = 1$, and take conditional expectations:

$$\mu_2 \equiv E(y \mid x = 0) = \alpha + E(u \mid x = 0), \tag{2}$$
$$\mu_1 \equiv E(y \mid x = 1) = \alpha + \beta + E(u \mid x = 1). \tag{3}$$

- Imposing Assumption (A3) yields:

$$E(y \mid x = 0) = \alpha + 0, \qquad E(y \mid x = 1) = \alpha + \beta + 0.$$

- Consequently,

$$\beta = E(y \mid x = 1) - E(y \mid x = 0) = \mu_1 - \mu_2. \tag{4}$$

Thus, β - **population raw differential**

Simple Regression when x is a Dummy

- Replacing population means by their sample averages, the OLS estimator is given by
$$\hat{\beta} = \bar{y}_1 - \bar{y}_2, \tag{5}$$
which is the **sample raw differential**. Moreover, $\hat{\alpha} = \bar{y}_2$.
- The fitted regression line $\hat{\alpha} + \hat{\beta} x_i$ can be written as
$$\hat{y}_i = \bar{y}_2 + (\bar{y}_1 - \bar{y}_2) x_i. \tag{6}$$
- The residual is defined as $\hat{u}_i \equiv y_i - \hat{y}_i$. Hence,
$$\hat{u}_i = \begin{cases} y_i - \bar{y}_2, & \text{if } x_i = 0; \\ y_i - \bar{y}_1, & \text{if } x_i = 1. \end{cases}$$

Simple Regression when x is a Dummy

Hypothesis test: To examine whether the population means of the two sub-samples are equal:

$$H_0 : \mu_1 = \mu_2 \quad \Longleftrightarrow \quad H_0 : \beta = 0.$$

- The test statistic is given by $\hat{\beta}/\mathrm{se}(\hat{\beta})$, which turns out to be exactly the same as

$$\frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\dfrac{\hat{\sigma}_1^2}{n_1} + \dfrac{\hat{\sigma}_2^2}{n_2}}}. \tag{7}$$

  This is a very well-known formula.

- One might ask why we need to estimate a regression model when it suffices to compute the two sample means and their variances to obtain the same result. The reason is that regression analysis provides a unified and general framework that encompasses a wide range of econometric models and hypothesis tests.

Simple Regression when x is a Dummy

- When the dependent variable is in logarithmic form,

$$\log y = \alpha + \beta x + u,$$

  then equation (5) becomes

$$\hat{\beta} = \log \bar{y}_1 - \log \bar{y}_2.$$

- The following algebra provides a unit-free interpretation:

$$\begin{aligned}
\hat{\beta} &= \log \bar{y}_1 - \log \bar{y}_2 \\
&= \log\left(1 + \frac{\bar{y}_1 - \bar{y}_2}{\bar{y}_2}\right) \\
&\equiv \log\left(1 + \frac{\%\Delta y}{100}\right) \\
&\approx \frac{\%\Delta y}{100}, \qquad \text{if } \frac{\%\Delta y}{100} \text{ is small.}
\end{aligned}$$

- Why does this approximation hold?
  $\log(1 + x) \approx x$ for small x (e.g., $|x| < 0.1$).

Simple Regression when x is a Dummy

- when $\%\Delta y/100$ is small, one can interpret $\hat{\beta}$ as the raw differential in the dependent variable. This is typically described as a **log-point change**.
- When $\%\Delta y/100$ is not small, the same concept must be computed from the third line:

$$\%\Delta y = 100\,[\exp(\hat{\beta}) - 1],$$

which is described as a **percentage-point raw differential**.

BHPS/US Example

- Formerly known as the British Household Panel Survey (BHPS).
- The BHPS/US is particularly well suited for studying differences between individuals since, by design, it records a rich set of information on people's lives.
- We will examine an extract from the data in order to replicate some of the examples discussed above.
- We will later show why panel data are superior to repeated cross-sections.
- The 14 variables that have been extracted correspond to 58,640 individuals i, who are observed one, two, or three times across waves $t = 1, 2, 3$.

BHPS/US Example

Data file: 20222_USoc_extract.dta

variable name	variable label
pidp	cross-wave person identifier
age	age
jbhrs	no. of hours normally worked per week
paygu	usual gross pay per month: current job
wave	wave
cpi	consumer price index (cpi)
year	year
region	Government Office Region
urate	regional unemployment rate
male	male dummy
race	ethnicity (5 categories)
educ	years of education
degree	degree, if any
mfsize9	firm size dummies (bin means)

BHPS/US Example

```
pidp:  280165, 541285, ..., 1.640e+09                    n =      58640
wave:  1, 2, ..., 3                                      T =          3
       Delta(wave) = 1 unit
       Span(wave)  = 3 periods
       (pidp*wave uniquely identifies each observation)


 Freq.  Percent    Cum. |  Pattern
---------------------------+---------
31091     53.02   53.02 |  111
10448     17.82   70.84 |  1..
 7178     12.24   83.08 |  11.
 3066      5.23   88.31 |  .11
 2855      4.87   93.18 |  ..1
 2206      3.76   96.94 |  1.1
 1796      3.06  100.00 |  .1.
---------------------------+---------
58640    100.00         |  XXX
```

There are 58,640 individuals observed a total of 133,272 times. The unit of observation is an individual–wave.

There are seven types of individuals: 31,091 are observed three times, . . ., and $10,448 + 2{,}855 + 1{,}796$ are observed only once.

BHPS/US Example

Some raw diffs

```
##
## ============================================================
##                              Dependent variable:
##                          -------------------------------------
##                                    lnhrpay
## ------------------------------------------------------------
## malemale                          0.183***
##                                    (0.005)
##
## Constant                          2.201***
##                                    (0.003)
##
## ------------------------------------------------------------
## Observations                       58,960
## R2                                  0.021
## Adjusted R2                         0.021
## Residual Std. Error          0.625 (df = 58958)
## F Statistic             1,251.147*** (df = 1; 58958)
## ============================================================
## Note:                          *p<0.1; **p<0.05; ***p<0.01
##                          Robust standard errors in parenthesis
```

Reference: R work `Regression_Intro_1.html`

BHPS/US Example

Some raw diffs

```
##
## =============================================================
##                              Dependent variable:
##                     ---------------------------------------
##                                    lnhrpay
## -------------------------------------------------------------
## gradany degree                   0.472***
##                                   (0.005)
##
## Constant                         2.138***
##                                   (0.003)
##
## -------------------------------------------------------------
## Observations                      58,942
## R2                                 0.119
## Adjusted R2                        0.119
## Residual Std. Error          0.592 (df = 58940)
## F Statistic             7,959.665*** (df = 1; 58940)
## =============================================================
## Note:                        *p<0.1; **p<0.05; ***p<0.01
##                     Robust standard errors in parenthesis
```

Reference: R work `Regression_Intro_1.html`

BHPS/US Example

- In `20222_USoc_extract.dta`, about half of the sample (59,216 observations) are in paid employment. Their ages vary substantially, and therefore they also differ considerably in their years of work experience, among other characteristics.
- When we take logs of real hourly pay and regress on a male dummy variable, we obtain an estimate of 0.183(0.0051). This says, on average earn0.183 log-points more than females which converts to 20.1%.
- A large proportion of individuals do not hold a university degree (102,644 observations), while the remainder do (30,397 observations). (Note the recoding of the degree variable.) When we regress log real hourly pay on a degree dummy variable, we obtain 0.472 (0.0051), which corresponds to 60.3%.
- This represents a very large raw differential. What is its implication over a lifetime?

Outlook

Next week: Lecture 4 Endogeneity, Lecture 5 Multiple regression

- In multiple regression, one can run both regressions simultaneously; that is, more than one right-hand-side (RHS) variable (in addition to the constant) can be included. One typically obtains similar, though not identical, estimates and standard errors.
- There are additional variables in `20222_USoc_extract.dta` relating to individuals' ethnicity, place of residence, and the size of the firm that employs them. These are multi-category dummy variables. Multiple regression applies in this context as well, although some recoding is required beforehand.
- The differences-in-differences methodology can also be implemented using multiple regression (covered in Lecture 7).
- Indeed, multiple regression is the workhorse technique in almost all of microeconometrics.